

DEVELOPMENT OF ENHANCED DEEP LEARNING ALGORITHMS FOR RCC GRADING FROM HISTOPATHOLOGY IMAGES

A Lightweight CNN Approach for Kidney Cancer Grading

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Introduction

- Renal Cell Carcinoma (RCC) is the most common form of kidney cancer, accounting for ~90% of cases
- It can metastasize to the lungs, bones, or brain if not detected early
- Diagnosis relies on grading histopathology images to determine severity
- Manual grading is time-consuming and depends heavily on expert interpretation

Problem Statement & Motivation

Problem Statement

Existing deep learning models for RCC grading from histopathology images are either accurate but computationally heavy, or lightweight but less accurate.

Motivation

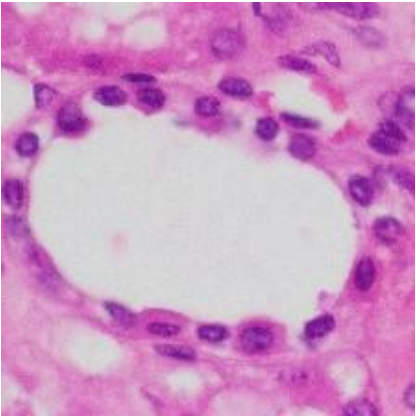
Design a lightweight CNN, optimized for the KMC Kidney Dataset, that improves classification accuracy while reducing computational complexity.

Major Contributions

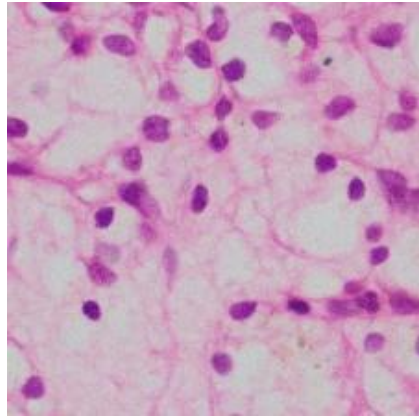
- Developed a custom CNN, built on a LiverNet backbone, integrating five specialized blocks: MK, RCAB, RPAB, MKRC, and SE-RFB
- Achieved 93.47% accuracy with only 410,725 parameters on the KMC Kidney Dataset
- Conducted a comparative study against 6 existing deep learning models, showing improved accuracy and efficiency

Dataset Overview

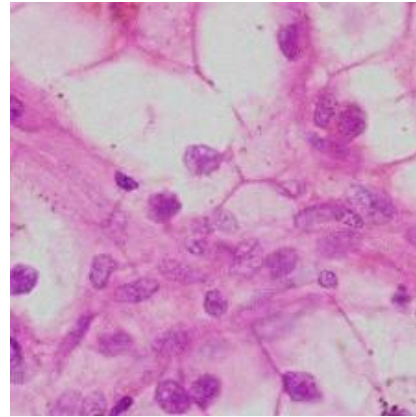
- KMC (Kasturba Medical College, Mangalore) Kidney Dataset
- 5,607 histopathology images across 5 grades (Grade 0 – Grade 4)
- Image format: 224 × 224



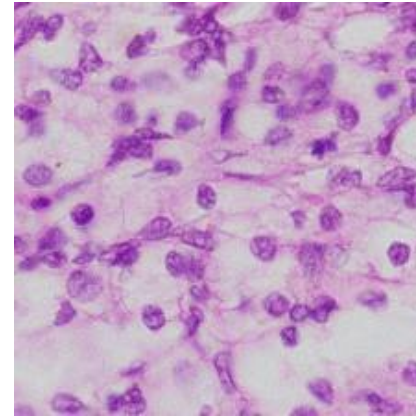
Grade 0



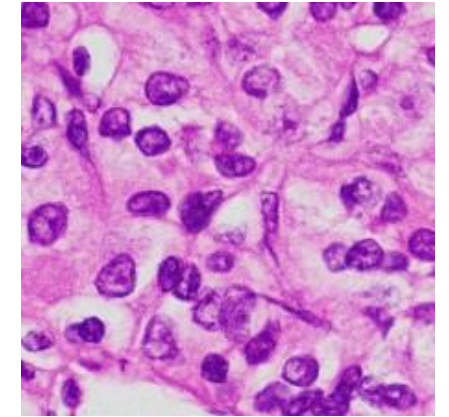
Grade 1



Grade 2



Grade 3

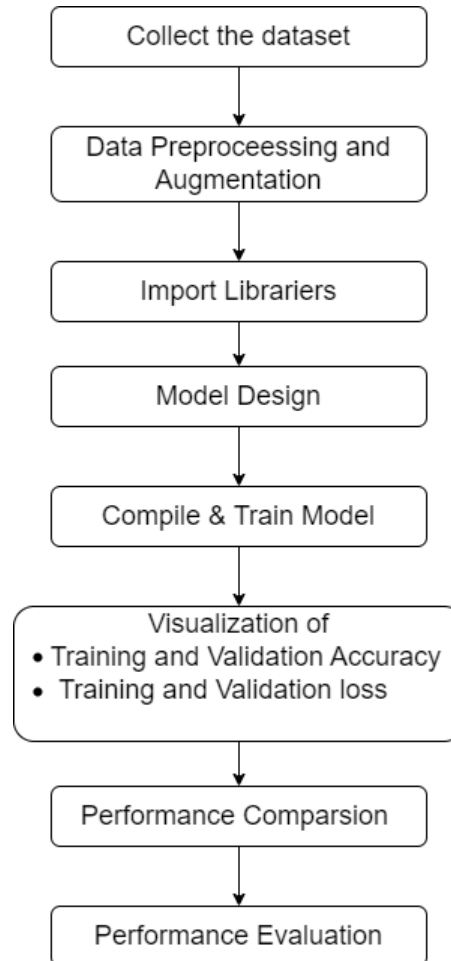


Grade 4

Sample images from the KMC Kidney Dataset (Grades 0–4)

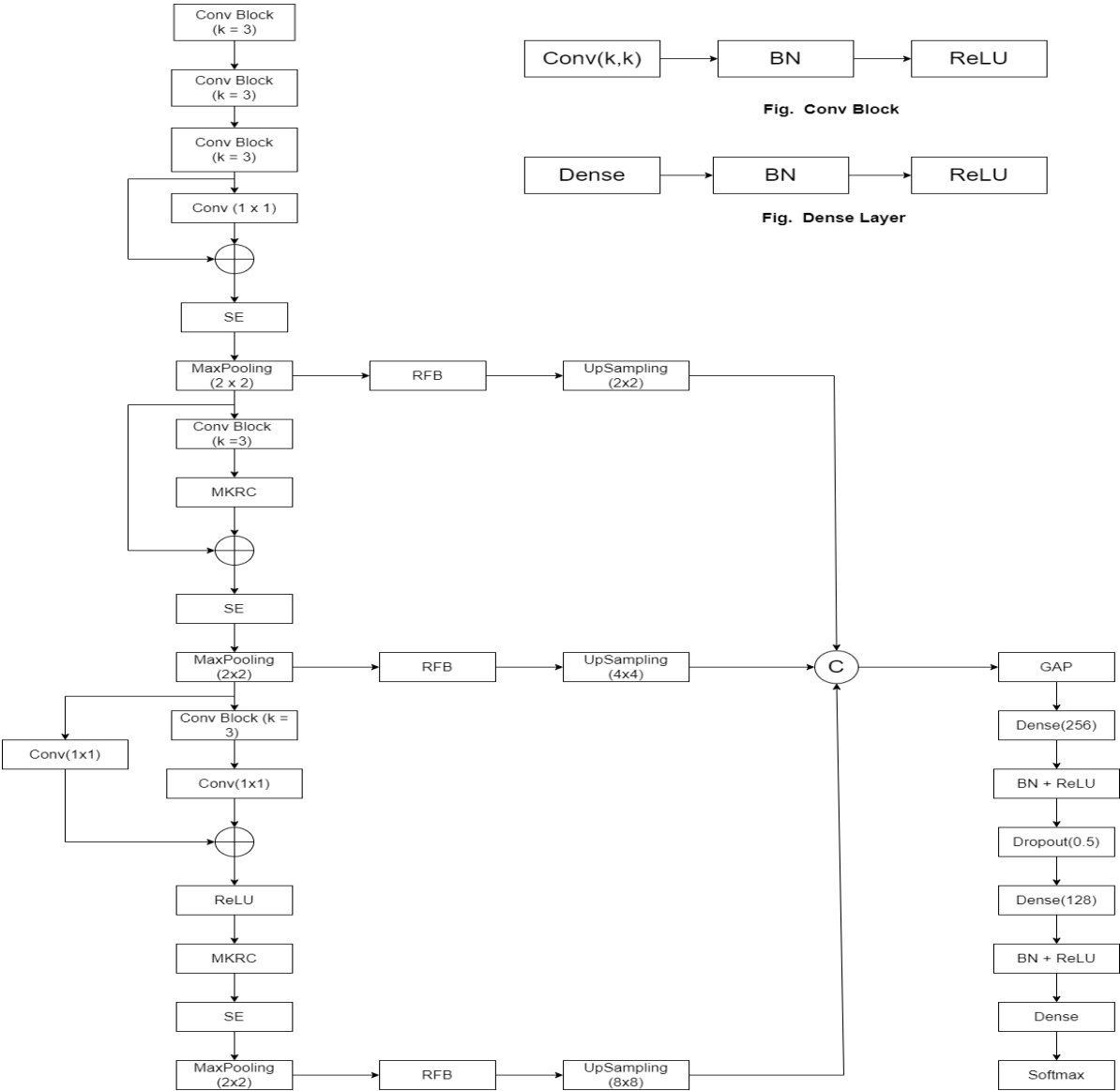
Methodology

End-to-end pipeline from data collection to performance evaluation



Proposed CNN Architecture

Built on a LiverNet backbone, extended with custom multi-kernel and attention blocks



Architecture Building Blocks — Multi-Kernel (MK) Block

- Applies parallel convolutions of kernel sizes 1×1, 3×3, 5×5 and 7×7
- Captures features at multiple scales in a single block
- Improves feature richness before deeper attention stages

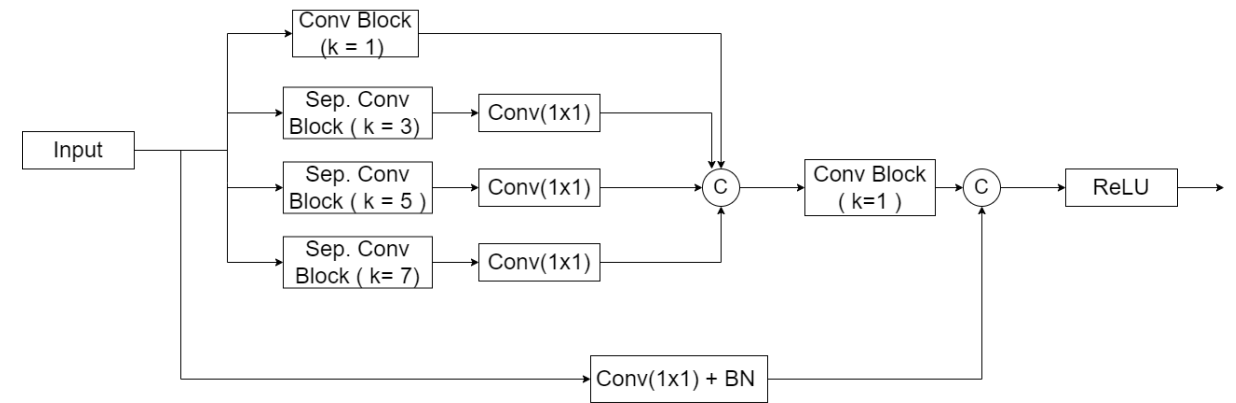


Figure: MK Block

Architecture Building Blocks — Attention Blocks

- RCAB (Residual Channel Attention Block): refines channel-level attention using global average pooling
- RPAB (Residual Pixel Attention Block): refines spatial (pixel-level) attention

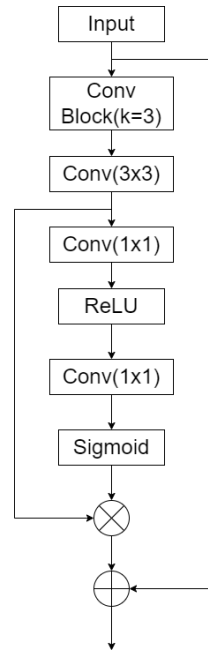


Figure: RCAB Block

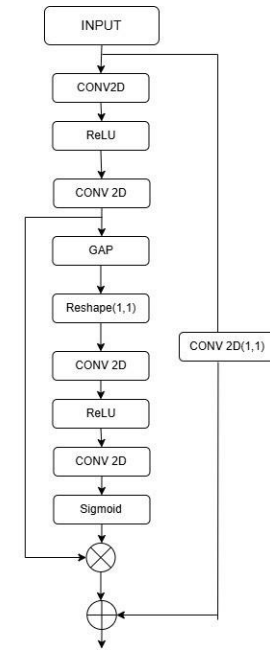


Figure: RPAB Block

Architecture Building Blocks — MKRC & SE-RFB

- MKRC (Multi-Kernel Residual Convolution): combines multi-kernel convolution with residual learning and channel attention
- SE-RFB (Squeeze-and-Excitation Receptive Field Block): enlarges the receptive field using contextual squeeze-and-excitation

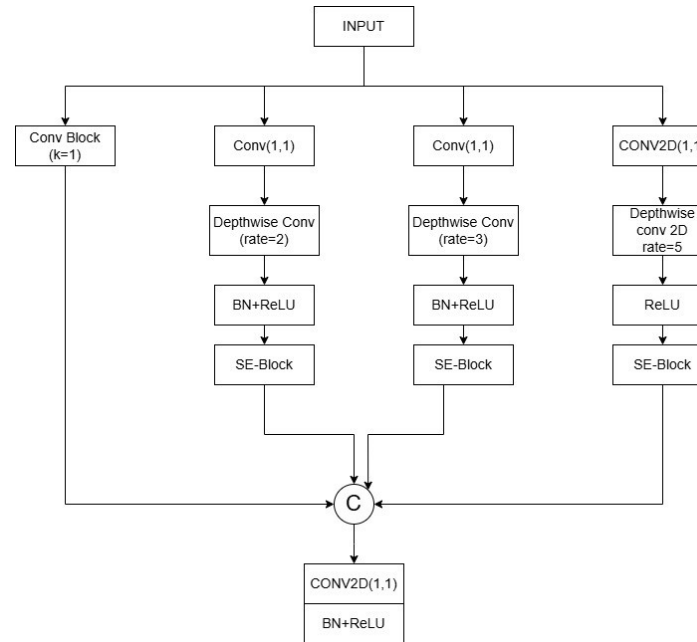


Figure: SE-RFB Block

Model Configuration

Block	Kernel Size(s)	Filters	Padding	Strides
Initial Conv Block	3×3	16–32	Same	1
MK Block	1×1, 3×3, 5×5, 7×7	64–128	Same	1
RPAB Block	3×3, 1×1	64	Same	1
RCAB Block	3×3 + 1×1 + GAP	64	Same	1
MKRC Block	1×1, 3×3, 5×5, 7×7	32–64	Same	1
SE-RFB Block	1×1 + DW-Conv (d=2,3,5)	32–64	Same	1

Results — Class-wise Performance

Grade	Accuracy	Precision	Recall	F1 Score	Jaccard Index
Grade 0	1.00	1.00	1.00	1.00	1.00
Grade 1	0.9763	0.899	1.00	0.947	0.899
Grade 2	0.9525	1.00	0.750	0.852	0.757
Grade 3	0.9584	0.833	1.00	0.904	0.833
Grade 4	0.9822	1.00	0.906	0.951	0.906
Overall	0.9347	0.9464	0.9312	0.9327	0.8777

Comparative Evaluation vs. Existing Models

Overall performance on the KMC Kidney Dataset across all grades

Model	Precision	Recall	F1-score	Accuracy
ShuffleNet (2018)	0.8340	0.8370	0.8332	0.8359
LiverNet (2021)	0.8999	0.8978	0.8985	0.8992
RCCGNet (2023)	0.8938	0.8908	0.8910	0.8932
TurkerNet (2024)	0.8892	0.8875	0.8874	0.8893
ProsGradNet (2025)	0.8962	0.8789	0.8800	0.8833
PBN (2025)	0.9032	0.9014	0.9016	0.9031
Proposed Model	0.9464	0.9312	0.9377	0.9347

Summary

- Proposed model achieved 93.47% overall accuracy on the KMC Kidney Dataset, outperforming all 6 compared models
- Delivered strong precision, recall, and F1-score with balanced performance across all grades
- Especially strong results for Grade 0 classification (perfect scores across all metrics)
- Achieved this with only 410,725 parameters — lightweight and deployable

Thank You